

Change in Global Temperature
relative to 1951-1980

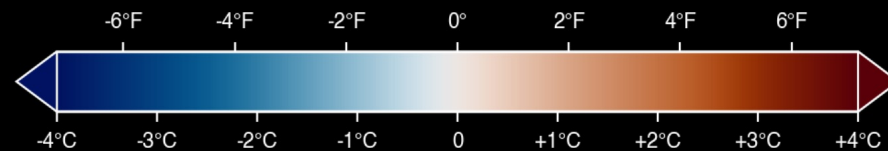
Where is Global Warming?

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NASA CCS Summer School

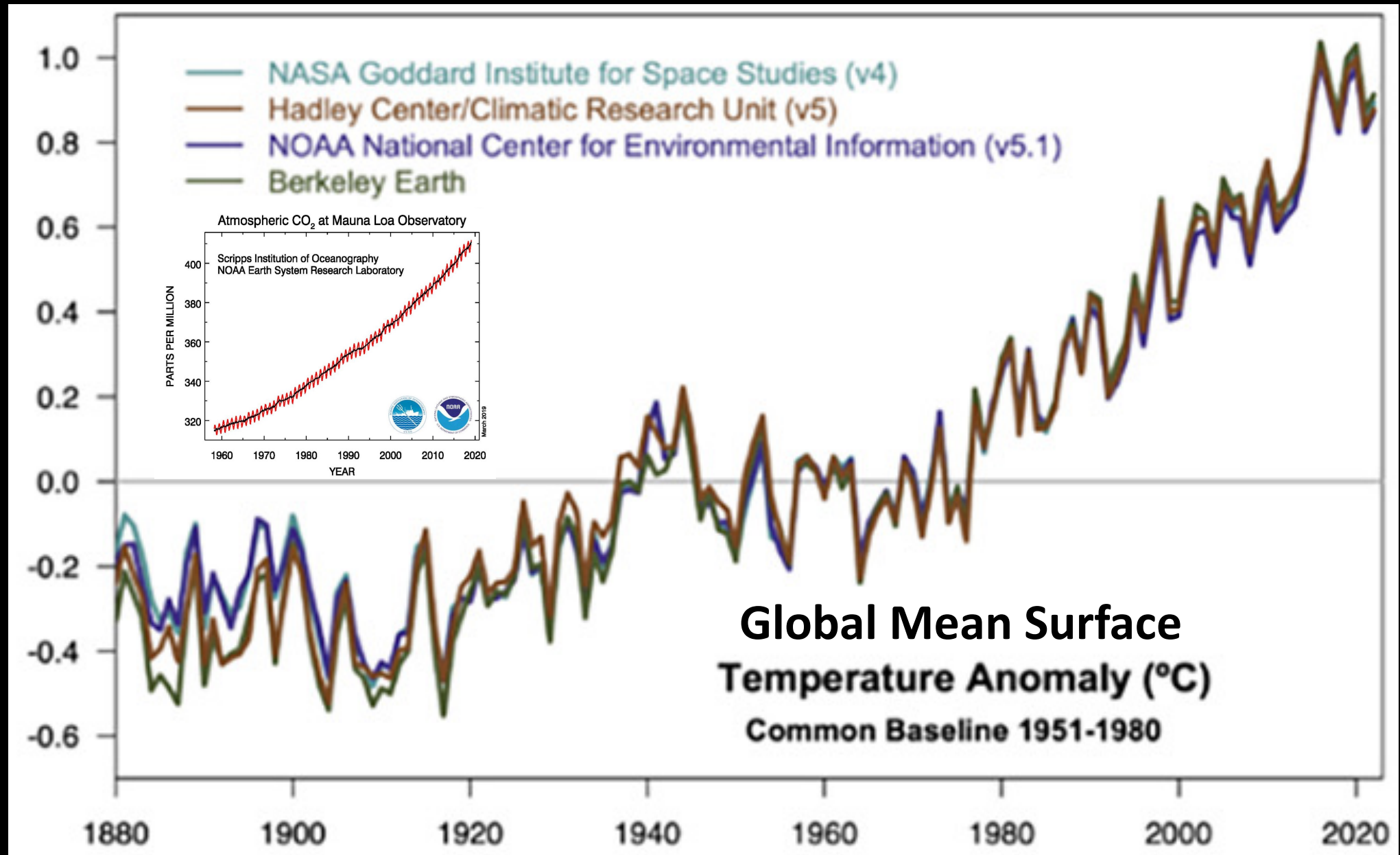
July 2026



GISS Surface Temperature Analysis
(GISTEMP v4)

Acknowledgements: Jinbo Wang, and Severine Fournier

Earth is warming



July 2023, the warmest month on record

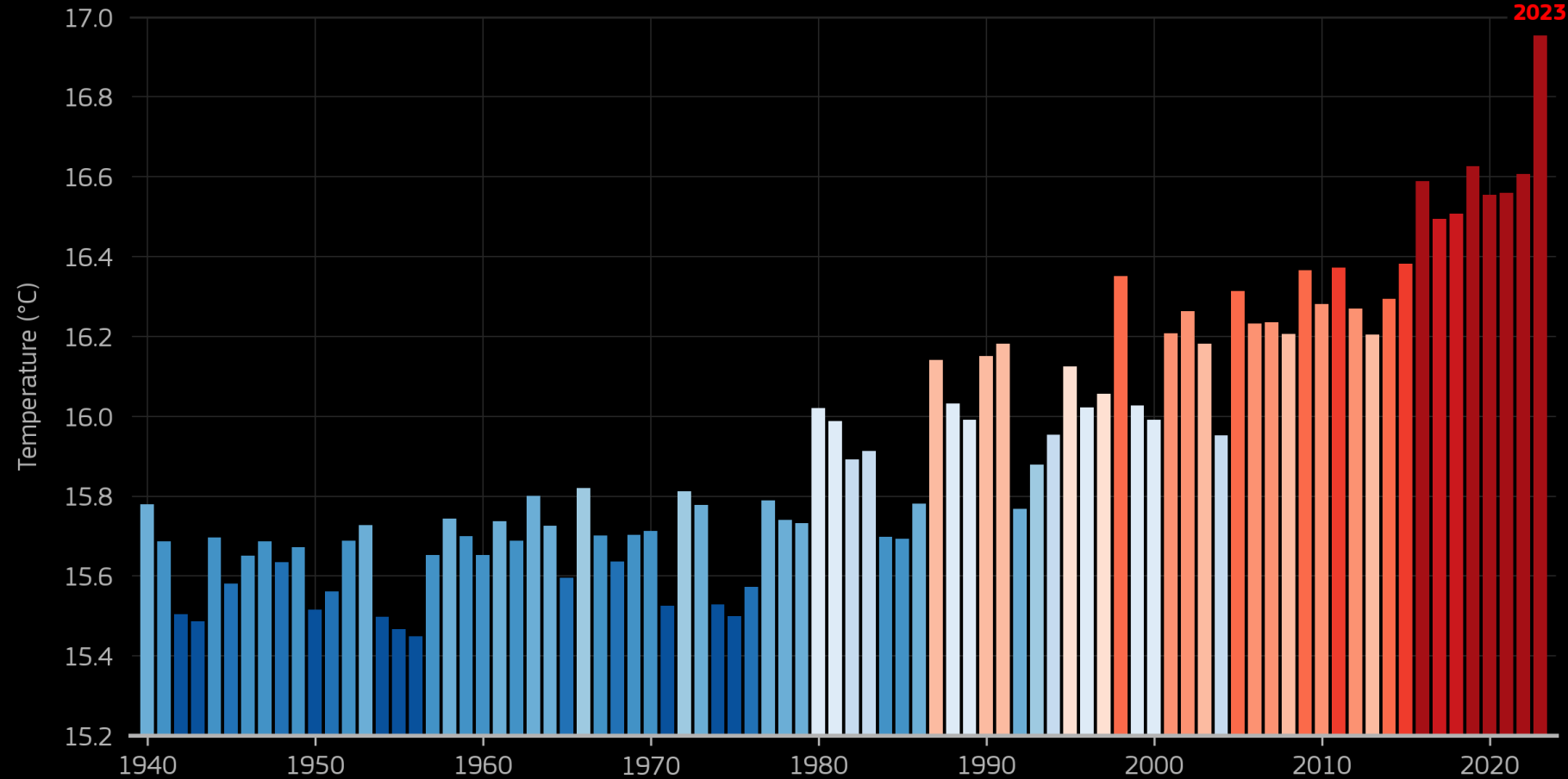
GLOBAL SURFACE AIR TEMPERATURE • JULY

Data: ERA5 1940–2023 • Credit: C3S/ECMWF



Climate
Change Service

climate.copernicus.eu



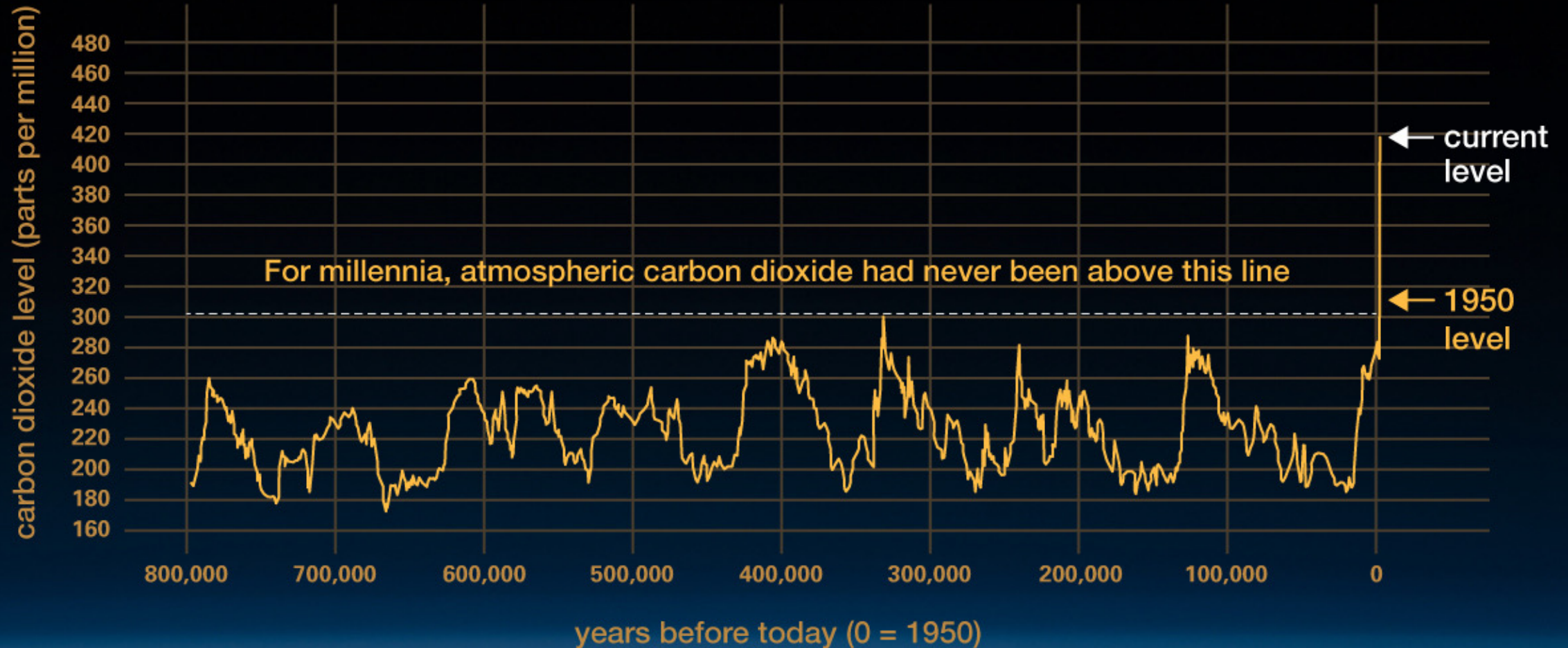
PROGRAMME OF
THE EUROPEAN UNION



Credit: C3S/ECMWF

What really causes global warming?

800,000 Years of Carbon Dioxide

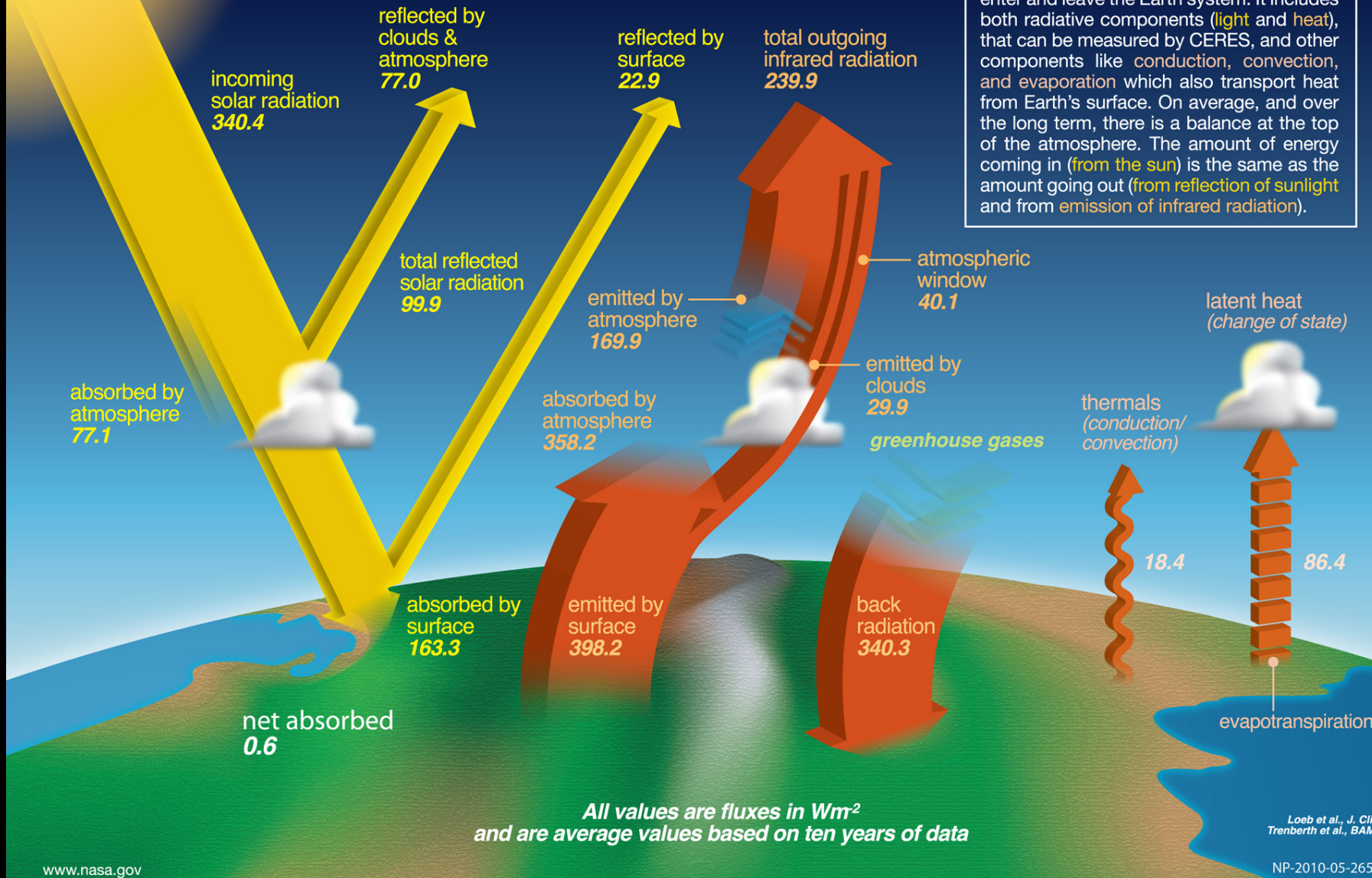




earth's energy *budget*

The Earth's energy budget describes the various kinds and amounts of energy that enter and leave the Earth system. It includes both radiative components (**light** and **heat**), that can be measured by CERES, and other components like conduction, convection, and evaporation which also transport heat from Earth's surface. On average, and over the long term, there is a balance at the top of the atmosphere. The amount of energy coming in (**from the sun**) is the same as the amount going out (**from reflection of sunlight** and from **emission of infrared radiation**).

~1 W/m² energy imbalance
(receiving more than emitting)



What happens to the heat?

Energy balance model

$$\rho C_p \frac{dT}{dt} = \text{Source} - \text{Sink}$$

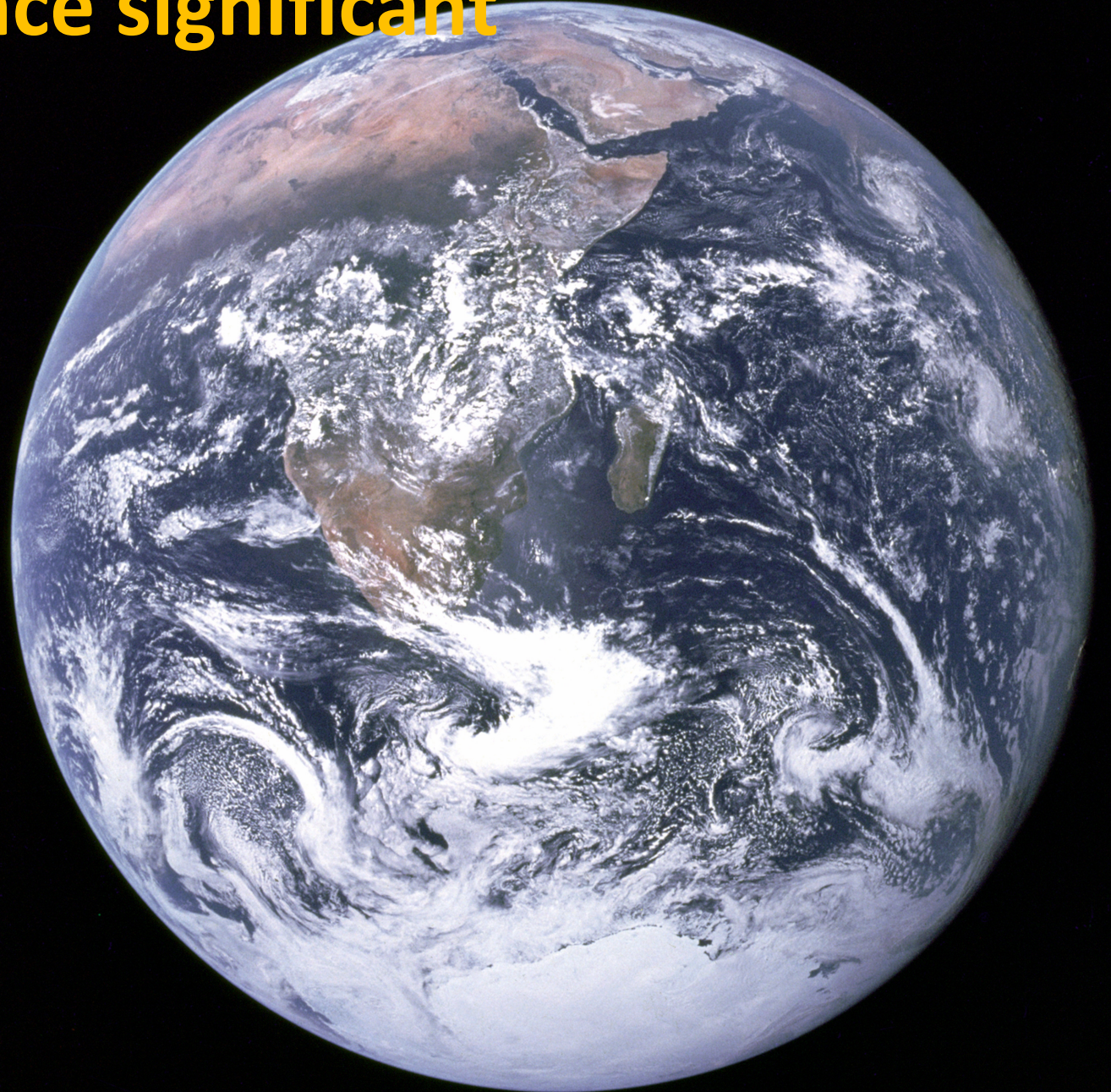


Is 1 W m^{-2} energy imbalance significant

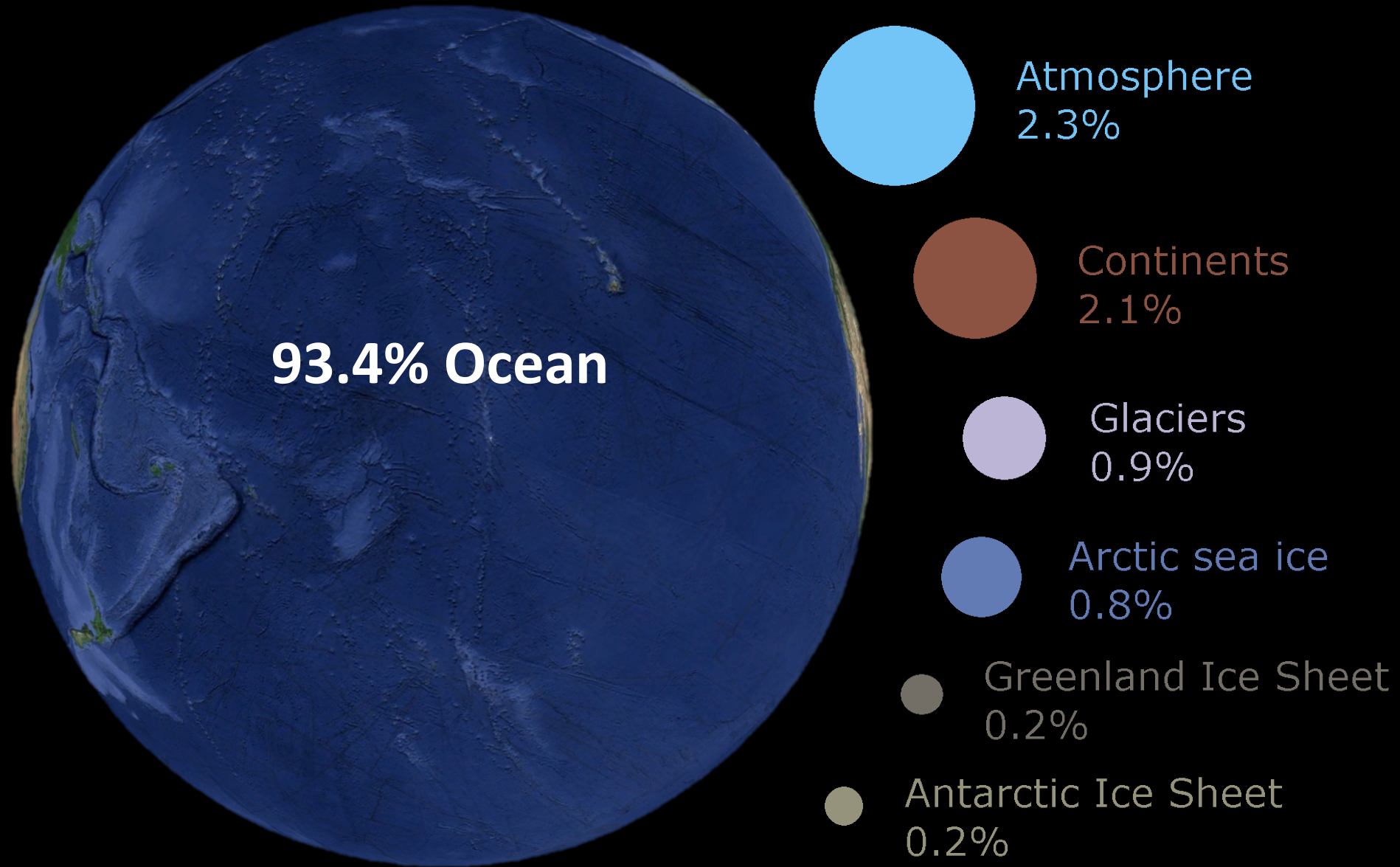
A $\sim 1 \text{ W/m}^2$ energy imbalance might seem small compared to the incoming solar radiation (340.4 W/m^2). But is it significant?

It certainly is, given the vast global surface area of Earth: $5.1 \times 10^{14} \text{ m}^2$.

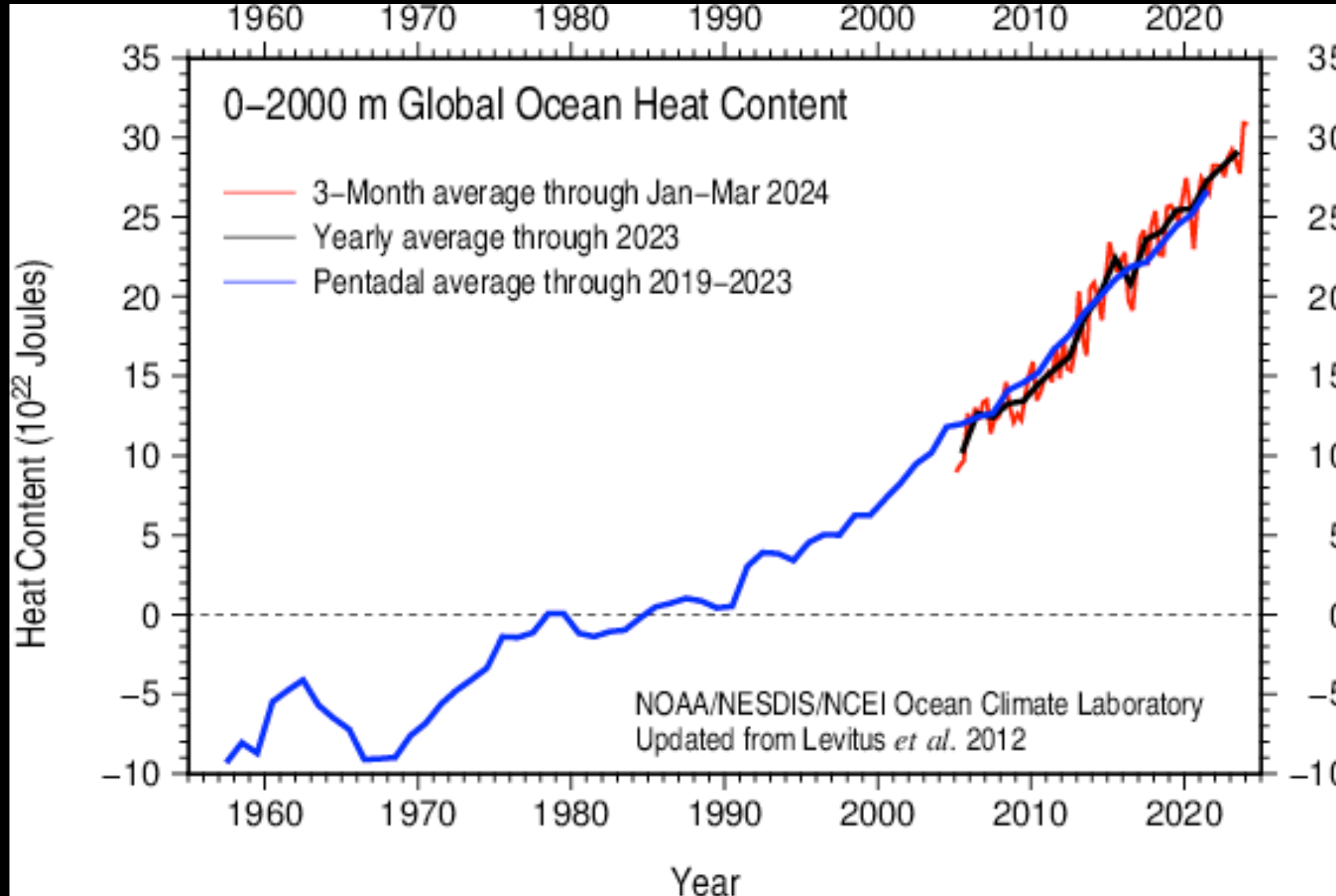
Each year, the global energy imbalance is more than 10 times the total energy consumed globally by humanity.



Where is all the heat going?



Ocean heat has steadily increased

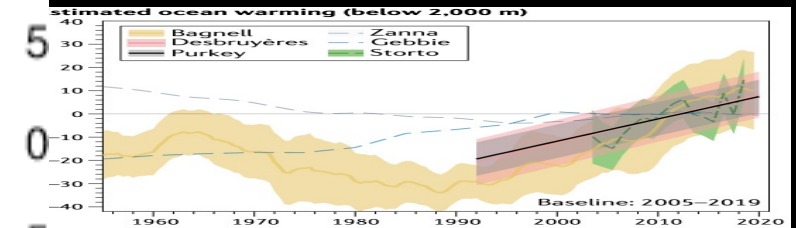


Global ocean heat content has steadily increased. The heat is mainly gained in the upper 2,000 m.

Since 2003, Argo floats have revolutionized the measurements of ocean properties at depth.

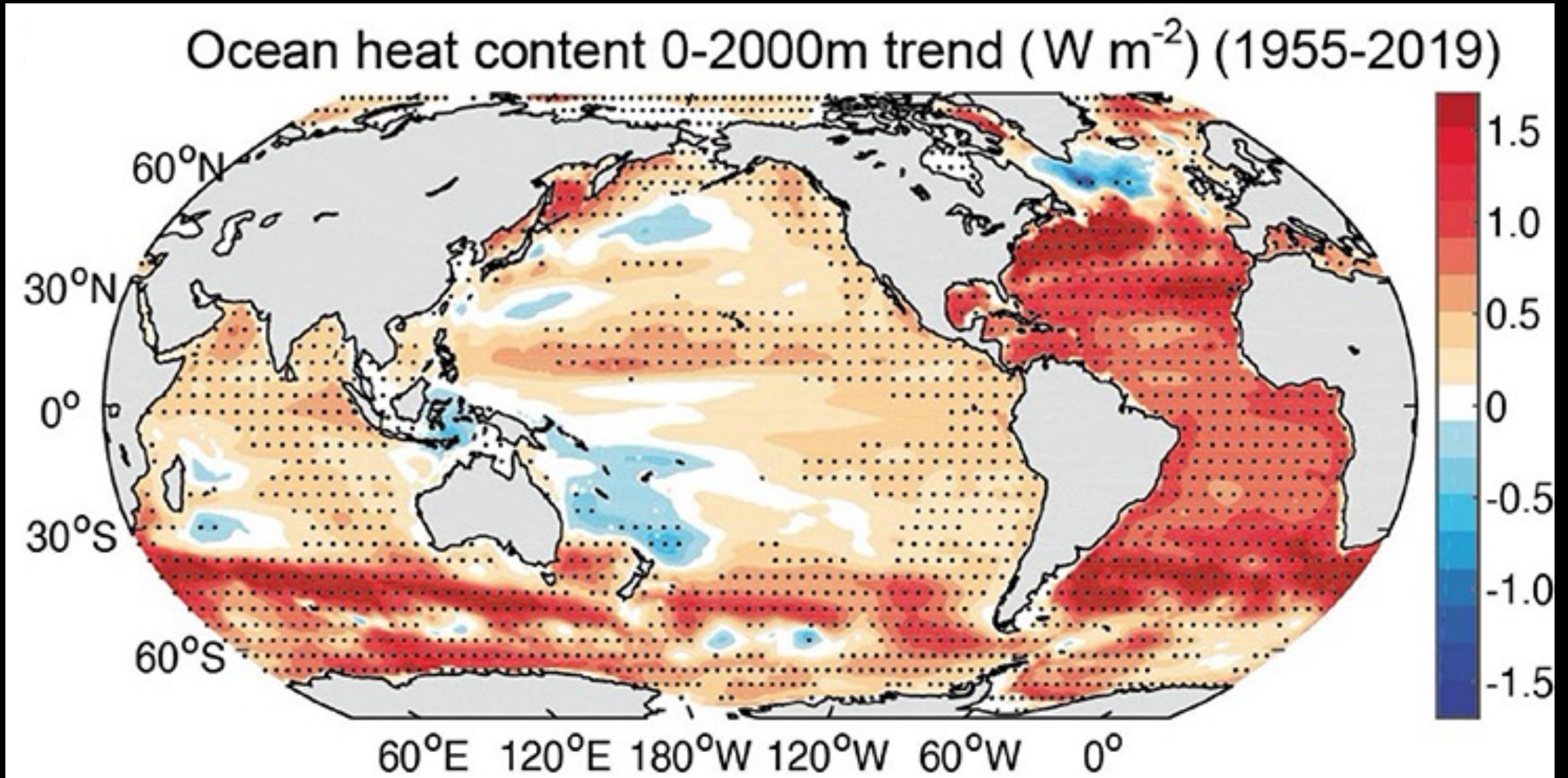


Below 2000 m

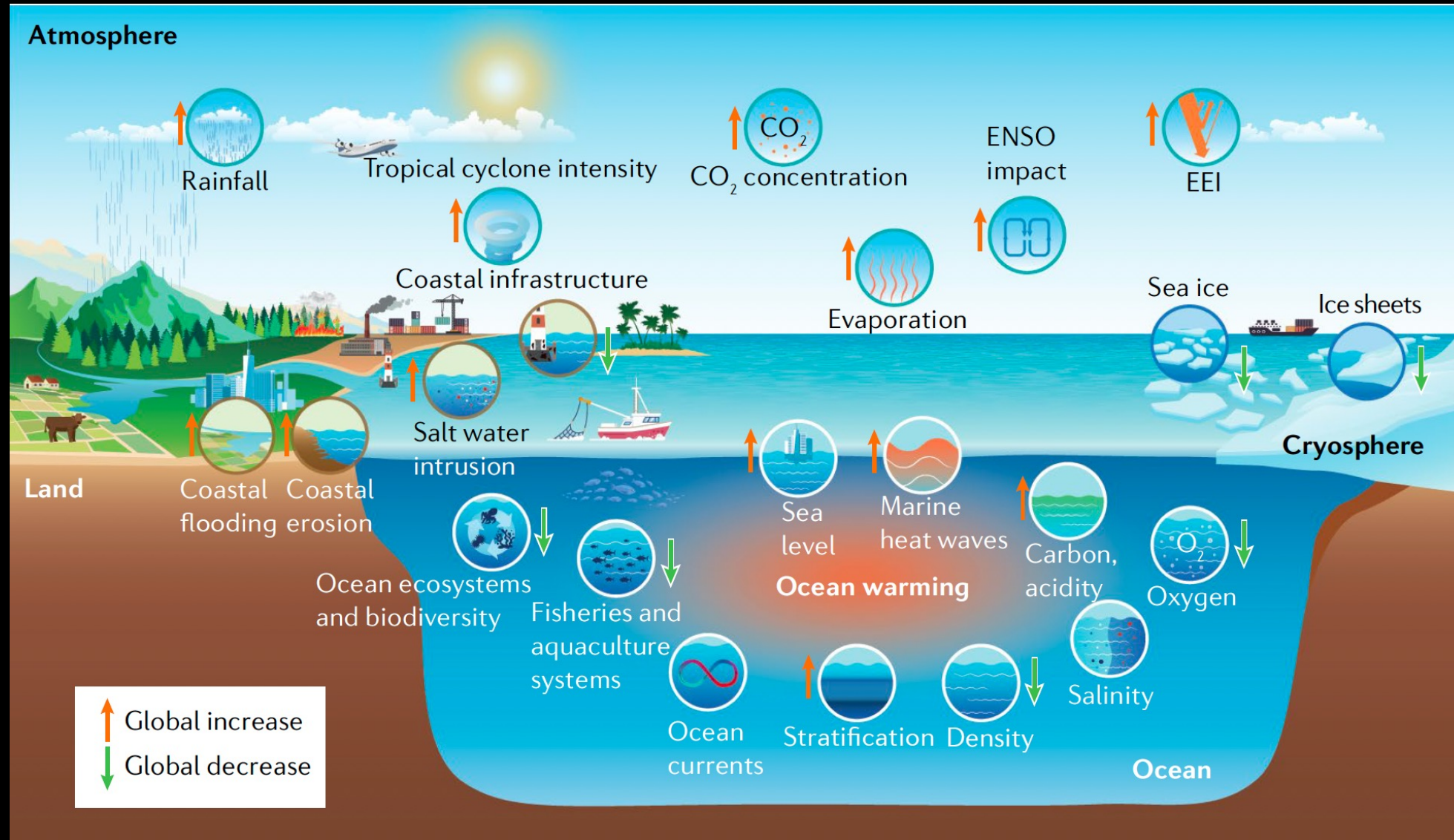


Cheng *et al.* (2022, *Nature*)

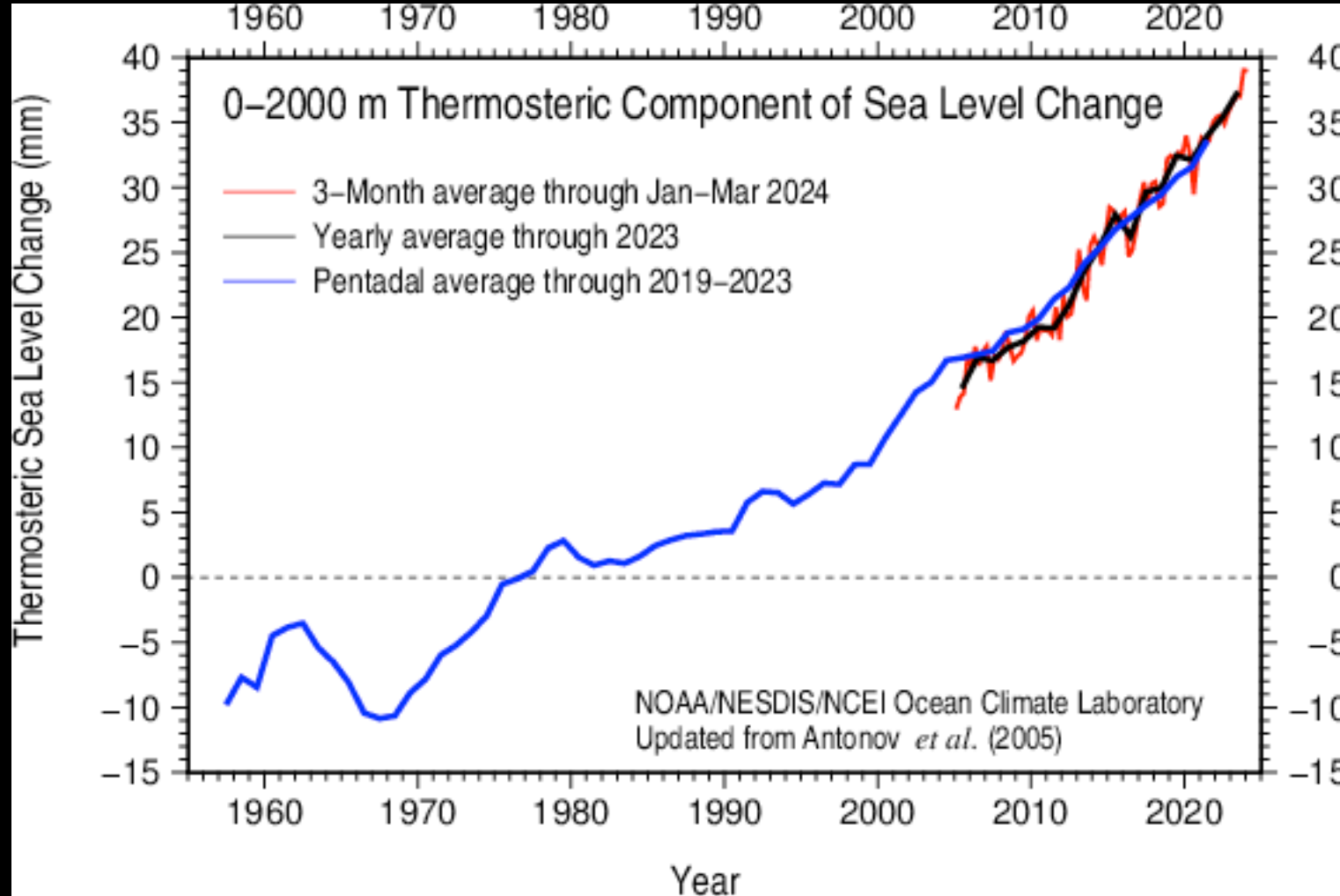
Ocean warming is not uniform



Linkage between ocean warming and changes in Earth system



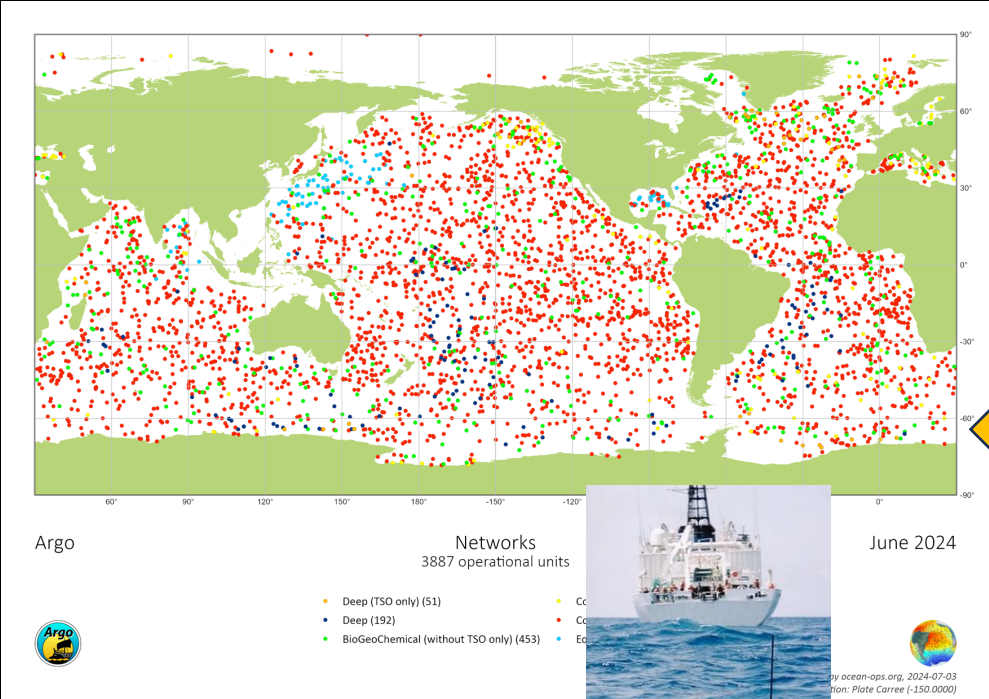
Sea Level Rise, a consequence of global warming



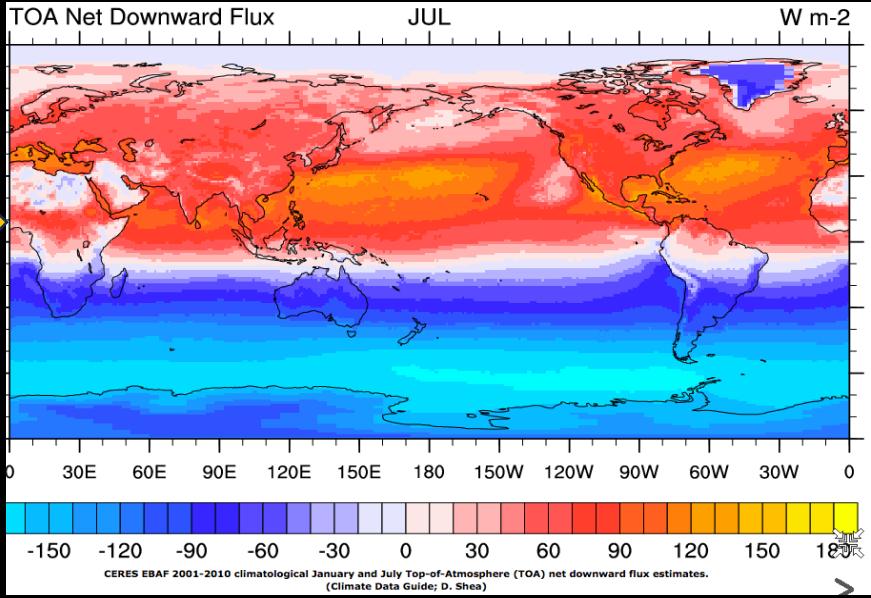
Since 1960: 5.5 inches global sea level rise:

1. The expansion of seawater associated with the warming ocean (thermal expansion) contributed about 1.8 inches (1/3)
2. Melting of ice sheets and mountain glaciers have contributed about 3.7 inches (2/3)

How to quantify where the energy from Earth's net radiative imbalance is accumulating?

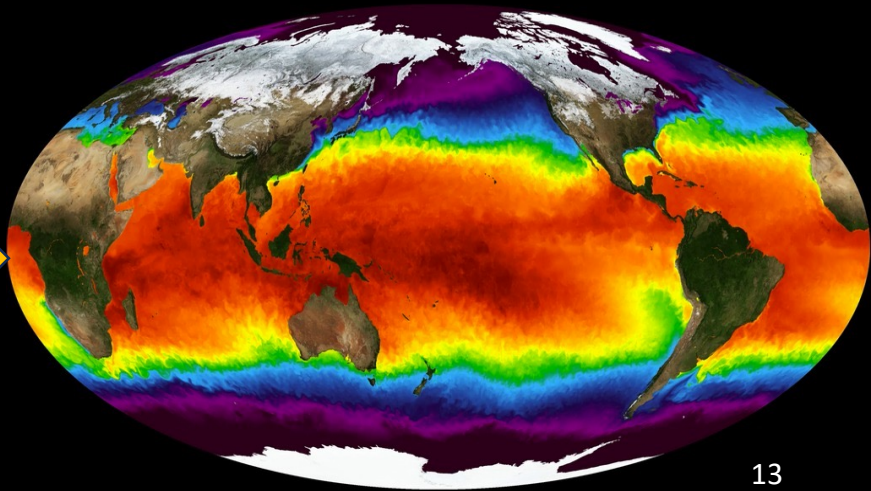


Top of atmosphere radiation
(Energy fluxes)



Argo global array of ocean profiling floats
(Ocean Interior Thermal Energy)

Satellite sea surface temperature
(Earth surface temperature)



Topic: Where is global warming?

- **Datasets:** CERES top-of-atmosphere shortwave and longwave radiation, Argo 3D ocean temperature, global sea level, and global ocean mass
- **Geographic foci:** mid latitudes (+60-30), low latitudes (+30-0)
- **Introduction:** In 2015 the Earth crossed a major global warming milestone: the global mean surface air temperature was one degree Celsius warmer than the mid-19th century pre-industrial average. Yet, the road to this 1 degree of atmospheric warming was bumpy: air temperatures in 2012 were the same as 1998 and 2002. If we really want to see where the excess energy of global warming is going, we need to quantify changes in the energy storage in Earth's largest thermal reservoir: **the ocean**.
- **Bonus:** Global mean sea levels are rising because of two main factors: ocean thermal expansion (from seawater warming) and the addition of liquid mass (from melting land ice). Today, the ocean observing system allows us to 'close' the sea level budget, in other words to compare total observed sea level rise from satellite altimetry with the expected sea level rise from ocean warming (using Argo temperature data), and the measured increase in ocean mass. With the datasets provided, you can compare sea level rise is associated with ocean warming and where it is associated with ocean mass increase.

Contact Scientists: Dr. Ou Wang (Ou.Wang@jpl.nasa.gov) and Ian Fenty (Ian.Fenty@jpl.nasa.gov)

Topic: Where is global warming?

- **Questions:**

1. Global warming is a result of an energy imbalance: more shortwave radiation absorbed at the top of the atmosphere (TOA) than re-emitted longwave and reflected shortwave. Calculate the global net radiative flux imbalance at the top of the atmosphere (TOA) and compare with time series from the latest CERES-EBAF data product. How does this compare with published estimates? How has this number changed through time? *Note: Note: when calculating TOA fluxes from gridded fields, account for grid cell area variations with latitude.*
2. If Earth's radiative flux imbalance were entirely absorbed in the troposphere (assume the lower 10 km of atmosphere), how fast would the troposphere warm each year on average? How does your predicted temperature change compare to the actual change through time? *Note: consider that atmospheric density decreases with height*
3. Repeat all parts of question (2) but instead consider distributing the entire radiative flux imbalance into the upper 10 m, 100 m, 700 m, and 2000 m of the global ocean, respectively. Compare the predicted temperature changes of the ocean against observed changes on a global scale. How do the actual warming trends of each of these depth categories compare against your predictions?
4. Divide the ocean into 4 basins: Pacific, Indian, Atlantic, and Southern Oceans. Which basins and which depths account for the greatest observed warming?

- **Notebooks:**

- **Meet the data notebooks are complete and re-runnable. The other notebooks are placeholders, ready for you.**

0. Project Introduction

1a-d: Meet the data: CERES Top of Atmosphere Fluxes, Argo ocean Temperature, Satellite Sea Level Anomaly , and GRACE Liquid Water Equivalent

2: TOA Energy Imbalance (CERES data)

3. Ocean Heat Uptake (Argo data)

4. Ocean Heat Uptake vs TAO (results from 2&3)

5. Sea Level Budget (uses Sea Level, Ocean Mass, and Argo temperature)

6. Regional Patterns (results from 1-5)

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